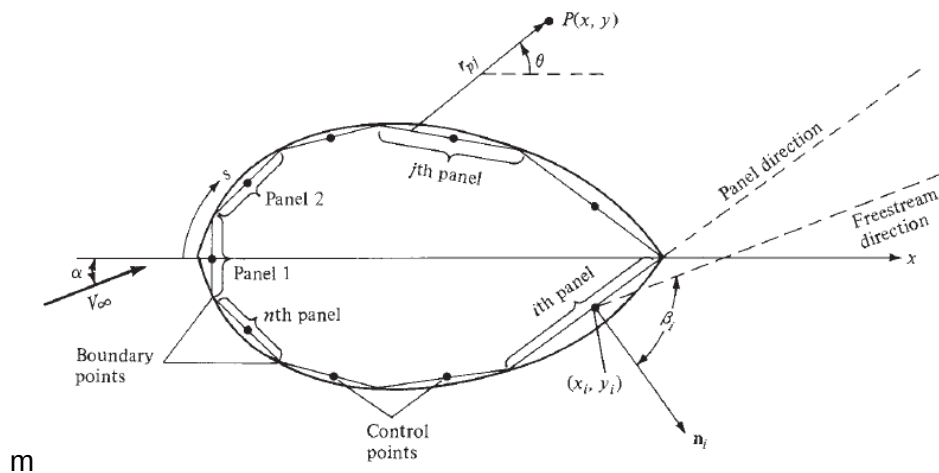


Panel Methods – Source Panel Method

- ➔ Panel methods are a way of discretizing arbitrary shapes and applying the distribution of elemental flows over the curved or un-curved surface to recreate the streamline patterns that look like the boundary (profile or surface of the object).
 - Source sheet for non-lifting bodies.
 - Vortex sheet for lifting bodies.
- ➔ Numerical methods must be used to solve the equations.
- ➔ Elemental flows helped us to solve specific flows like semi-infinite bodies, Rankine oval, Stationary cylinders, and Rotating cylinders.
- ➔ But if the shape is arbitrary like airfoils (symmetric or cambered), wedges, and fuselage, then we need to use panel methods.
- ➔ With panel methods, we discretize complex aerodynamic shapes into panels.
- ➔ On each panel, we place a mini elemental flow. By tuning the strength of each flow element, we can recreate the streamlines that represent the original aerodynamic shape.
- ➔ Because of the discretization, this technique is primarily computational.

Discretization:

- First consider an airfoil. Generally, the surface of the foil represents a continuous smooth curve.
- Divide the curve into many small **connected** segments called **Panels**.

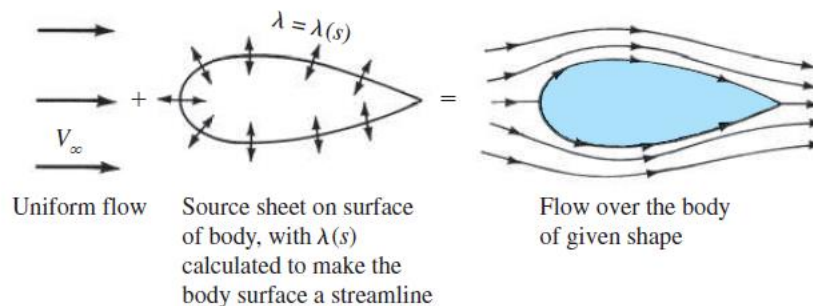


- The ends of the segments are called boundary points.
- The center of the segments is called the control points.
- Each segment (panel) gets a number that rises sequentially.

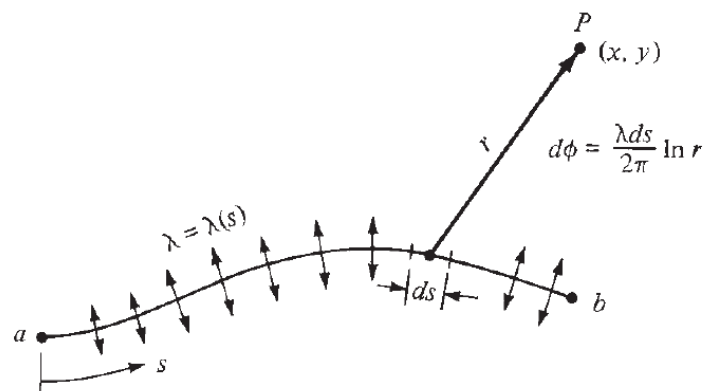
- Keep in mind that the surface of the foil does not actually exist and the panel does not represent the solid surface.
- Elemental flows are placed on the panels. (Source $\rightarrow$ Source Panel Method or Vortex $\rightarrow$ Vortex Panel Method).

Source Panel Method:

- This is a method that recreates the shape of a body using a source sheet or a series of closely spaced sources or sinks.
- Source Panel Method is purely for **non-lifting bodies**. There is **no addition of circulation** through these flow elements, therefore **no lift**.
- This limits the application to primarily symmetric bodies with no angle of attack.



- Consider a source sheet along a curve. The curve s starts at point A and ends at point B. The sheet strength is depicted as λ .



- $\lambda = \lambda(s)$, source strength per unit length along s .
- The strength of a single source is represented by Λ which represents the **volumetric flow rate per unit depth into the page**.
- However, we are dealing with a series of elements along a curve, and we consider the strength in segments along that curve.
- $\lambda \rightarrow$ Volumetric flow rate per unit depth and unit length s .

- We are interested in how a segment of the curve ds acts on the point p . The distance between the point p and the segment is r .
- The velocity potential at point p due to the segment ds is

$$d\phi = \frac{\lambda ds}{2\pi} \ln r$$

$\lambda ds \rightarrow$ Strength of source sheet of a small segment. λ is a sheet strength as a function of a point on the curve.

Recall for a source,

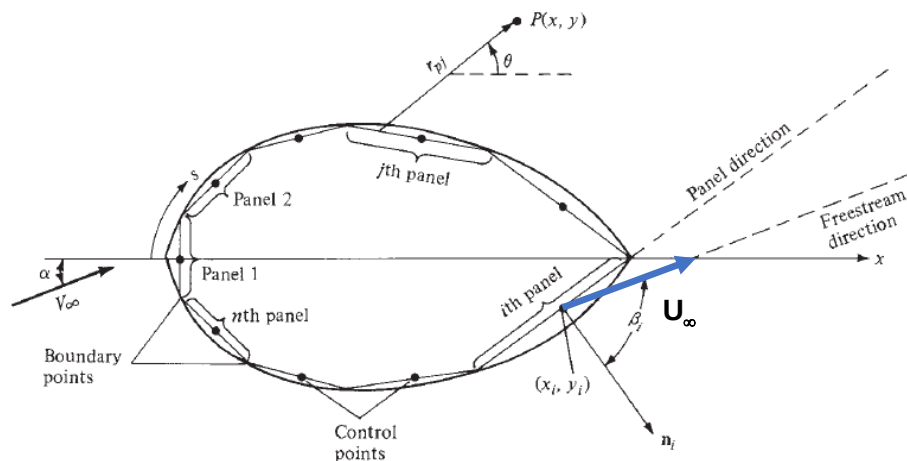
$$\phi = \frac{\Lambda}{2\pi} \ln r$$

- The complete velocity potential at point p due to all the segments,

$$\phi(x, y) = \int_a^b \frac{\lambda ds}{2\pi} \ln r$$

If the λ is right, then the streamline field accurately reflects the body shape that we are interested.

- Develop the equation to solve for the sheet strength distribution (λ) such that the streamline represents the body.
- Let us approximate the source sheet by a series of straight panels. To accomplish this and to work with the closed surface we need to discretize the sheet into panels.



- Considering the point p , the potential of p from the j^{th} panel alone

$$\Delta\phi_j = \frac{\lambda_j}{2\pi} \int_j \ln r_{pj} ds_j$$

Velocity potential induced at point p due to the j^{th} panel.

The strength parameter is constant across a segment. So, it is outside the integral.

- The velocity potential at point p from all the panels,

$$\phi(p) = \sum_{j=1}^n \Delta\phi_j = \sum_{j=1}^n \frac{\lambda_j}{2\pi} \int_j \ln r_{pj} ds_j$$

This equation successfully defines the velocity potential throughout the entire flow field because the point p can be chosen anywhere we want. However, this equation is unsolvable because λ is still unknown.

- The distance r_{pj} is given by,

$$r_{pj} = \sqrt{(x - x_j)^2 + (y - y_j)^2}$$

Where (x_j, y_j) are coordinates along the surface of the j^{th} panel.

- Since point P is just an arbitrary point in the flow, let us put P at the control point of the i^{th} panel.

$$\phi(x_i, y_i) = \sum_{j=1}^n \Delta\phi_j = \sum_{j=1}^n \frac{\lambda_j}{2\pi} \int_j \ln r_{ij} ds_j$$

$$\text{Where } r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Boundary condition:

- Flow cannot go through the surface or penetrate it. Velocity normal to the wall is zero (at the wall).
- Consider the i^{th} panel, the normal velocity on the i^{th} panel has two specific contributions.
 → **Due to freestream:** A portion of the flow is normal to the surface. The angle between the freestream and the unit normal vector of i^{th} panel is β_i .

$$U_{\infty, n} = U_{\infty} \cdot \mathbf{n} = U_{\infty} \cos \beta_i$$

$U_{\infty, n}$ is positive when directed away from the surface.

- **Due to surrounding panels:** Normal component of velocity induced by surrounding panels.

$$U_n = \frac{\partial}{\partial n_i} [\phi(x_i, y_i)]$$

- Two contributions to the normal velocity due to freestream and the surrounding panels get added together and set equal to zero, because of the surface boundary condition.

$$\frac{\lambda_i}{2} + \sum_{j=1}^n \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_i} (\ln r_{ij} ds_j) + U_{\infty} \cos \beta_i = 0$$

This is a linear algebraic equation with n unknowns $(\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n)$.

- The first term $\lambda_i/2$ is the normal velocity induced at the i th control point by the i th panel itself.
 - Consequently, a singular point arises on the i th panel because when $j = i$, at the control point itself $r_{ij} = 0$.
 - It can be shown that when $j = i$, the contribution to the derivative is simply $\lambda_i/2$.
- The second term is the summation is the normal velocity induced at the i th control point by all the other panels.
- The third term is the normal velocity induced at the i th control point due to freestream velocity.
- Let s be the distance along the body surface. The component of freestream velocity tangent to the surface is

$$U_{\infty,s} = U_{\infty} \sin \beta_i$$

- The tangential velocity at the control point of the i th panel induced by all the panels is obtained by

$$U_s = \frac{\partial}{\partial s} [\phi(x_i, y_i)]$$

- The total surface velocity at the i th panel is

$$U_i = U_{\infty,s} + U_s = \sum_{j=1}^n \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial s} (\ln r_{ij} ds_j) + U_{\infty} \sin \beta_i$$

- In turn, the pressure coefficient at the i th control point is obtained from

$$C_p = 1 - (U_i/U_{\infty})^2$$

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.